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(COURSE DESCRIPTION)

DATE: 2026/09/10
INDEX: 27013

Karaj Girls Technical and Vocational College - Karaj

Discontinuous Bachelor
Computer Software Technology Engineering

Remarks:
According to the Academic Instruction of University, in the academic system, each theoretical credit is presented in 16 or 17 hours, each practical credit in 32 or 34 hours,each of the workshop credit in 48 or 51 (except Professional Certificates) and each internship or training credit in 64 to 120 hours.

There are some different type of the courses passing by a student at this university, as listed below:

1. Basic
2. Core / Major
3. Specialized
4. Elective / Optional
5. General
6. Internship / Workshop
7. Project / Thesis

Each one of these types contain some courses to be passed, according to the educational regulations passed by the Ministry of Science, Research and Technology or Ministry of Health & Medical Education of Iran.

The grading system in this university is from 0 to 20. The minimum passing grade for a course leading to an Associate's Degree or a Bachelor's Degree is 10, for a course leading to a Master's Degree and Medical Sciences is 12 and for a course leading to a PhD Degree is 14.

Note: This document has been prepared at **Zahra Koosha's** request in the **Course Description** Platform. Scan the QR-code below to certify this document.

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University Info :

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Course	Credits	Description
Differential Equations	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Basic)** In this course, first and second levels of linear differential equations and some nonlinear differential equations will be introduced, in addition, students will learn about some numerical and analytical ways to solve Mathematical Problems. **Syllabus :** Nature of differential equations and their solution, family of graphs and vertical routes, physical patterns, separable equation, first order linear differential equation, homogeneous equation, second order linear equation, homogeneous equation with fixed constants, method of indefinite constants, method of changing parameters, application of second order equations in physics and mechanics, solution of differential equation with series, Bessel and Gamma functions, Legendre polynomial, an introduction to differential equations set, Laplace transform and its application in solving differential equations.
Engineering Mathematics	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Basic)** Familiarization with the concepts and applications of series and Fourier transform, fields of occurrence and methods of analysis of equations with partial derivatives, functions and mappings in the complex plane. **Syllabus :** Fourier series and Fourier transform: definition of Fourier series, Euler's formula of half-period expansion, forced functions, Fourier integral, Laplace transform; Gibbs theorem and limits of Fourier analysis; orthogonal and complete functions; series convergence and Fourier transform; application of Fourier analysis in solving engineering problems; symmetric properties of transformation and Fourier series; Introduction of short time Fourier transform. partial equations or derivatives: one-variable wave equation; method of separation of variables; Dalembert's solution for the wave equation; Heat diffusion equation; wave equation; Laplace's equation in Cartesian, spherical and polar coordinates; Elliptic, parabolic and hyperbolic equations; Application cases of Laplace transform in solving equations with partial derivatives; Solving partial derivative equations using the Fourier integral; Derivation of telegraphic equations. Analytical functions and isometric mapping and complex integrals: limit and continuity; derivative of complex functions; exponential and trigonometric functions; hyperbolic and logarithmic; Inverse and exponential trigonometry with mixed view; maintaining the Dirichlet boundary condition; Conservation of energy under isometric transformation; Application of isometric transformation in electric capacity calculations. Line integral in the complex plane: Cauchy's integral theorem; Calculating the linear integral using indefinite integrals; Cauchy formula; Taylor and McLaren expansions; Calculation of integrals by the method of balances; The remaining case; Calculation of the real integral of the fraction; Calculation of real trigonometric integral.

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Discrete Mathematics	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Basic)** This course covers elementary discrete mathematics. It emphasizes mathematical definitions and proofs as well as applicable methods. Topics include formal logic notation, proof methods; induction, well-ordering; sets, relations; elementary graph theory; integer congruences; asymptotic notation and growth of functions; permutations and combinations, and counting principles.
Engineering Probability & Statistics	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Basic)** Acquiring the ability to analyze engineering issues in the field of forecasts and planning in order to use it to make decisions. **Syllabus :** The role of probability statistics in various fields of engineering. Basic concepts of probability. Events and their probability in set theory and mathematical expressions. Analytical models to express random phenomena. Random variables, normal distributions, probabilities. Functions of random variables. Probability distribution, functions of one and several variables, mean and standard deviation, probability functions. Estimation of required parameters using available data. Regression and correlation analysis. Basic formulas and their application. Application of ellipse method in engineering sciences.
Artificial intelligence	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** AI is an introductory course in Artificial Intelligence. The goal is to acquire knowledge on intelligent systems and agents, formalization of knowledge, reasoning with and without uncertainty, machine learning and applications at a basic level. **Syllabus :** Problem solving, reasoning, planning, natural language understanding, computer vision, automatic programming, machine learning, and so on. Of course, these topics are closely related with each other. For example, the knowledge acquired through learning can be used both for problem solving and for reasoning. In fact, the skill for problem solving itself should be acquired through learning. Also, methods for problem solving are useful both for reasoning and planning. Further, both natural language understanding and computer vision can be solved using methods developed in the field of pattern recognition.

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Computer Graphics 1	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** The course aims to provide students with a solid foundation in computer graphics principles and techniques, including 2D and 3D graphics, rendering, animation, and interactive systems. **Syllabus :** Introduction to computer graphics. Graphics pipeline and hardware. Digital image representation. Color models. Geometric transformations. Viewing and projection. Curves and surfaces. Lighting and shading models. Texture mapping. Ray tracing. Graphics programming with OpenGL. Animation techniques. Virtual reality concepts. Advanced topics in computer graphics.
Internet Engineering	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** This course provides an introduction to fundamental concepts of the Internet and web technologies including their architecture, protocols and applications. The main topics of the course include: Introduction to Internet history and Internet services, networking basics, overview of TCP/IP (addressing, routing and transport protocols), network programming, web programming, web services, web servers and Internet security.
Computer Simulation	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** This course covers the principles and concepts of computer simulation. Simulation implements the behavior of a system at the level of detail required in a virtual environment. This makes it possible to study systems with low cost and high flexibility. Simulation applications cover a wide range, from training in the simulated environment to evaluating system designs and setting parameters that affect the performance of a system. In this course, students are introduced to the principles of implementing and building simulators and implement a simulation tool using a programming language. In addition, a standard simulation package is examined and how to perform the simulation and the language of the relevant software in that environment are explained. This course also describes the statistical methods used to perform the simulation, including how to model the input and analyze the output, as well as the validation of the simulation model. Methods of reducing variance and simulating rare events are also examined.
Special Topics	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Mandatory)** Teaching and discussing a new topic in the field of study with the opinion of the faculty and the supervisor.

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Software Project	3	**(Duration: 96 hours Practical)** **(Type of Course: Specialized - Mandatory)** student choose a professor of computer science department and he/she defines a subject such as implementing a program on special scientific usage or a subject in computer software to research about that, and student must provide an essay and submit their project result and present it.
Internship	3	**(Duration: 96 hours Practical)** **(Type of Course: Specialized - Mandatory)** Students must practically get involved in industry and market by working totally 240 hours in an appropriate and defined company in either governmental or private sectors as an intern.
Network operating system	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** This course introduces operating system concepts for single-user systems. Topics include hardware management, file and memory management, system configuration/optimization, and utilities. Upon completion, students should be able to perform operating systems functions at the support level in a single-user environment.
Network operating system workshop	2	**(Duration: 64 hours Practical)** **(Type of Course: Specialized - Optional)** In this lab exercises, students will work with virtual machines to install and configure different network operating systems. They will also work with network services and practice troubleshooting techniques.
Network installation	2	**(Duration: 64 hours Practical)** **(Type of Course: Specialized - Optional)** The Computer Network Installation and Maintenance course covers management, maintenance, and troubleshooting of a network. On the course you learn the skills necessary to begin working immediately in the computer and network technical support business.
Web Page Design	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** **Syllabus :** First Section: Introduction to Home page, comparison of home page with another methods of presentation of information, standards, importance of information selection, use of menu, form, image, font, data update, link between pages, get information from users, Second Section: HTML application and its structure, instructions for HEAD, BODY, COMMENT, lists, menus, tables, graphic files, images, image maps, form, table design, familiarity with front page, familiarity with some important web servers, web server roles, NCSA, introduction to java and java script and its capabilities

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Object-Oriented System Design	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Specialized - Optional)** **Syllabus :** Concepts of this course is by C++ and one project must be presented. Introduction to C++, IO in C++, constructor and destructor functions, encapsulation, inheritance, structure class, union, in-line functions, object allocation, object transferring into functions, friend functions, recursive functions, arrays, pointers and reference pointers, overload functions, binary and logical and relational operators, multiple inheritance, IO systems, IO mani pulators, extractor, inserters, random access, IO control, customize, derived classes, virtual classes, polymorphism and its application.
Algorithms Design	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Core)** The main purpose of this course is to teach the basic concepts and common methods of analysis and design of algorithms. In this lesson, students learn how to analyze a given problem and find some possible algorithms to solve it. Then analyze and compare those algorithms in terms of computational complexity and identify the best ones based on the characteristics of the input samples. In this lesson, basic algorithms for solving some common problems will be introduced. **Syllabus :** Review of essential points of Data Structures, problem solving methods (for each methods some problems and special algorithm for the problem shall be represented and being analyzed), introducing to complexity, divide & conquer method (problems to be analyzed: max and min of an array, multiplication of two n-bit number, Strassen method about matrix multiplication, round robin algorithm, sorting with Quicksort algorithm), dynamic programming method (matrix multiplication, traveling sales man problem, polygon triangulation),greedy algorithms(scheduling problem, Huffman Code, making change),methods based one xhaustive search, alpha-beta pruning (puzzle, tic-tac-tac), revelation of methods for problems, graph algorithms(searching methods of graph, Dijkstra algorithms, minimum spanning tree, Floyd algorithm, topological sorting and...), maximum flow networks and other problems.

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Machine and Assembly Languages	2	**(Duration: 32 hours Practical & 16 hours Theoretical)** **(Type of Course: Core)** Reminding use of different units of computer, definition of register, introduction of its types and its application, concept of instruction format in machine language, types of instructions, different stages of instruction execution, methods of addressing (implied , immediate, direct, indirect, indexing, base, relocation, related to content of program counter, paging).Concept of relocation, instruction by registers , instruction by memory (manipulation of half word full word, bytes and bits), jump instructions and control of loop , logical instructions, dexinal manipulation , floating point manipulation subprograms, recursive subprograms, macros, trace and debugging of program, damping introduce assembly and its types.
Software Engineering	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Core)** The purpose of this course is to address the engineering points that must be observed in all stages of software production. In this course, students are introduced to the activities and tools needed to produce a software product. **Syllabus :** Importance and objectives of software engineering; planning; stages and design stages controlling; principles of needs analysis; methods of needs analysis; principles of software design; study of several known methods of design such as data structure, data flow, functional, aimed and immediate analysis, outstanding points of programming languages in viewpoint of software engineering, software accuracy and confidentiality, software maintenance, performance of a practical project.
Software Engineering Lab	1	**(Duration: 32 hours Practical)** **(Type of Course: Core)** This course is offered to undergraduate students and its purpose is to address the applied topics of software engineering. In this lesson, engineering methods are experienced in the form of practical projects. The work of the laboratory is carried out in five main areas of software engineering, including requirements engineering, analysis, design, implementation and testing. The laboratory consists of 10 three-hour sessions. Each group is assigned a project that will work on that project and its documentation until the end of the semester.

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Computer Architecture	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Core)** An introduction to computer configuration, introducing different generations of computer, register transfer and micro-operations, register transfer language, inter-register transfer, computational micro-operations, sliding micro-operations, control functions. **Syllabus :** Basic computer organization and its design, instruction codes, computer instructions, scheduling and controlling, designing a sample computer such as PDP/8 and its micro-operations, methods of numbers representation, representation by fixed point, representation by floating point, other binary codes, errors revealing codes, organization of central processor including systems with several processor registers and bass system and systems using stack, study of several computers such as PDP/11, IBM 370. Designing computational processors, comparison and subtraction of binary numbers without sign, algorithm of multiplication and division with fixed and floating point. -Input and output organization, memory organization, auxiliary memories, memory of microcomputer, hierarchy of memory, associative memory, dummy memory, cache memory, memory management hardware.
Computer Architecture Lab	1	**(Duration: 32 hours Practical)** **(Type of Course: Core)** The purpose of this laboratory is to acquaint students with the concepts presented in the computer architecture course such as designing digital circuits, how to analyze and debug them, design logic and arithmetic parts of a processor, design memory units, input / output, data path and control unit in a typical processor , Using hardware description languages such as Verilog and VHDL to describe their structure, synthesize and finally implement on FPGA boards. **Syllabus :** Familiarity with CAD tools to design and test the correct operation of logical circuits. Familiarity with a sample simulator (such as Quartus). Designing, implementing and testing the performance of a sample circuit (such as the circuit for the addition of two decimal numbers) using a simulator. Design and implementation of computing architectures. Design and implementation of 4-bit constant check multiplier. Design and implementation of floating point adder/subtractor. Design and implementation of decimal to binary converter. Designing and implementing the architecture of a simple computer. Design and implementation of the calculation unit with the possibility of choosing the origin. Design and implementation of the calculation unit with the possibility of control by the program. Designing and implementing a complete computer with data memory and jump instructions. Processor design and implementation. Design and implementation of microprogrammable control circuit. Performance test of the implemented circuit.

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System programming	3	**(Duration: 32 hours Practical & 32 hours Theoretical)** **(Type of Course: Core)** Executive files structure, viruses operation, programs trace, loaders, relocation, position independent code, call operating system functions, heap memory allocation, TSR programs, application of interrupts, disk partitioning and work with them by operating system functions and BIOS, file table structure, forms of file storage, work with ports, display text and picture on monitors with high resolution, familiarity with device drivers, familiarity with addressing in protected cases, virtual on 80X86 processors. In this course, concepts is presented by assembly programming language.
Scientific and Technical Materials Presentation Methods	2	**(Duration: 32 hours Theoretical)** **(Type of Course: Core)** **Syllabus :** Different types of scientific and technical subjects (letters, reports, pamphlets, manual and etc.), common points in all scientific and technical writings: specifying the objective of writing and its eventual readers, organizing the subjects, abstract of essay together with report, the role of a good introduction, dividing the subjects into parts and chapters, discussion and conclusion, preparing source and reference index, attachments, preparing the pictures and diagrams and tables. Important points in translation of scientific and technical subjects, writing style, marking and its importance, preparing final format of writing by typing machine or computer, foot-article, notes and other lateral subjects, an introduction to research methods, presenting subjects orally, effective use of audio-visual devices.
English for Software	3	**(Duration: 48 hours Theoretical)** **(Type of Course: Core)** In this course, while familiarizing students with general concepts and keywords used in the field of computer science and engineering, the methods and skills needed to study and write scientific and specialized texts in this field are taught. The main emphasis is on strengthening students' skills in reading and understanding specialized texts in the computer field, but work will also be done on improving students' writing skills and listening comprehension. The lesson materials are divided into three parts: main topics, optional topics, and auxiliary topics. In each three-hour session (including two hours of lessons and one hour of practice), two or three of the main topics are taught. Optional topics will be taught if there is time, but the usual practice is to give these topics as exercises. Auxiliary topics include studying new materials in the computer field and extracurricular activities (such as holding student presentations and using multimedia content).
Sport I	1	**(Duration: 32 hours Practical)** **(Type of Course: General)** This course is designed for the prospective coach, physical education and/or recreation professional.

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Islamic Thoughts 2 (Prophethood & Imamate)	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** This course aims to introduce the concepts of prophecy and Imamate in Islam, exploring their significance and principles in Islamic thought and theology.
Thematic Interpretation of Quran	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** Analyzing the thematic aspects of the Quran
Analytical history of early Islam	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** Familiarity of students with how Islam emerged and spread and the way of transition from ignorant society to Islamic society.
Islamic Revolution of Iran	2	**(Duration: 32 hours Theoretical)** **(Type of Course: General)** Theoretical acquaintance with the causes and factors of the emergence of the Islamic Revolution and the study of the cultural, social and political developments of the Islamic Revolution and the issues after it.

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